

Influence of hyperbaric oxygen on left ventricular contractility, total coronary blood flow, and myocardial oxygen consumption in the conscious dog

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Savitt MA, Rankin JS, Elberry JR, Owen CH, Camporesi EM. Influence of hyperbaric oxygen on left ventricular contractility, total coronary blood flow, and myocardial oxygen consumption in the conscious dog. *Undersea & Hyperbaric Med* 1994; 21(2):169-183.—It is known that hyperbaric oxygenation (HBO) decreases total coronary blood flow (TCBF) and cardiac output (CO). To determine whether this is related to an alteration in myocardial contractility, 10 chronically instrumented conscious dogs were studied during pharmacologic autonomic blockade. Left ventricular (LV) volume was measured with ultrasonic transducers, LV transmural pressure with micromanometers, TCBF with Doppler-flow probes, and coronary AVO₂ difference (A-CSO₂) was calculated from direct LV and coronary sinus (CS) sampling. To evaluate the effect of increased oxygenation, data were obtained during resting control conditions and during dynamic vena caval occlusions (VCO), at 1 atmosphere of pressure, while breathing air (1 bar/0.21); at 3 atmospheres, breathing compressed air (3 bar/0.21), and at 3 atmospheres breathing 100% oxygen (3 bar/1.0). Because of autonomic blockade, heart rate (HR) was not statistically different in the three conditions. With increasing oxygenation, arterial oxygen tension (Pa_{O₂}) increased from 85 ± 5 mmHg (mean ± SD) at 1 bar/0.21, to 1374 ± 201 mmHg at 3 bar/1.0 whereas arterial carbon dioxide tension (Pa_{CO₂}) and pH values were not statistically different. Arterial oxygen content (AO₂ content) and CSO₂ content increased significantly (both $P < 0.05$) with increasing Pa_{O₂}. LV stroke volume (SV), CO, coronary blood flow, and myocardial oxygen consumption (MVO₂) were all significantly reduced ($P < 0.05$) with increasing levels of oxygenation. Intrinsic myocardial function, as measured by the stroke-work/end-diastolic volume relationship was unchanged from 1 bar/0.21 to 3 bar/0.21, and to 3 bar/1.0 ($P < 0.20$). Thus, the diminished TCBF, CO, and MVO₂ associated with HBO do not seem to be associated with a primary alteration in myocardial contractility, but rather may result from a physiologic autoregulation of the myocardium to increasing levels of Pa_{O₂}.

hyperbaric oxygenation, Doppler-flow probe, coronary sinus saturation, coronary blood flow, myocardial oxygenation

Hyperbaric oxygen (HBO) therapy, during which a patient inhales 100% oxygen while being exposed to a pressure higher than sea level, is now utilized as an adjuvant in the treatment protocol for various oxygenation disorders. An increasing number of critically ill patients are being referred for hyperbaric therapy. Patients with gas gangrene, necrotizing fasciitis, severe air embolism, decompression sickness, and carbon monoxide and cyanide poisoning occasionally may exhibit extreme hemodynamic instability while undergoing HBO therapy for treatment of their disease. An evaluation of the therapeutic efficacy of HBO depends on a knowledge of both the clinical and physiologic responses of patients to HBO. With more critically ill patients receiving HBO, it is important for clinicians to understand what effects this therapy has on the cardiovascular system.

Extensive research has been performed in the areas of oxygen toxicity and the effects of HBO on neural and pulmonary tissues; however, much less is known about its influence on the cardiovascular system. It has been postulated that HBO may be an effective adjuvant in the treatment of coronary occlusive disorders because an increase in oxygen partial pressure to 15 times over air breathing at 1 atm can be achieved when 3 atm abs of oxygen are administered to patients. This concept is dependent on whether myocardial performance is maintained during exposure to HBO (1, 2). HBO has consistently been shown to produce a decrease in cardiac output (CO) and coronary blood flow (CBF) (1-3), while investigators continue to debate as to whether these reductions are a result of decreased myocardial contractility (3-5). Recent investigations have shown that left ventricular (LV) pump work and cardiac contractility are maintained or slightly elevated during exposure to HBO (4, 5).

Previous studies have been limited because of the semi-quantitative methods used to determine myocardial performance. The present study was designed to utilize accepted methods for defining myocardial performance in an attempt to understand the actual adaptations, if any, of the cardiovascular system to HBO. The data presented in this communication will attempt to resolve many previously conflicting hypotheses and explain the varied results obtained in the past, with the hope of better understanding the physiology of HBO exposure in the conscious dog.

METHODS

Experimental preparation

Ten healthy adult dogs (15-22 kg) were premedicated with procaine penicillin G (6×10^5 units), dihydrostreptomycin (250 mg), and iron dextran (100 mg). Each dog was anesthetized with sodium thiamylal (20 mg/kg i.v.), intubated, and ventilated with a volume respirator (Puritan-Bennett, MA2, Los Angeles, CA). Under sterile conditions, a left thoracotomy was performed through the fifth intercostal space. The pericardium was opened transversely and the heart was suspended in a pericardial sling. The heart was instrumented as shown in Fig. 1. Silastic pneumatic occluders were positioned around the inferior and superior vena cavae. Pulse-transit ultrasonic dimension transducers were positioned across the LV anterior-posterior (minor), base-apex (major), and septal-free wall diameters. Cuffed 20 MHz Doppler-flow probes (i.d. 3 mm, o.d. 5 mm; Craig Hartley, Baylor University Medical Center,

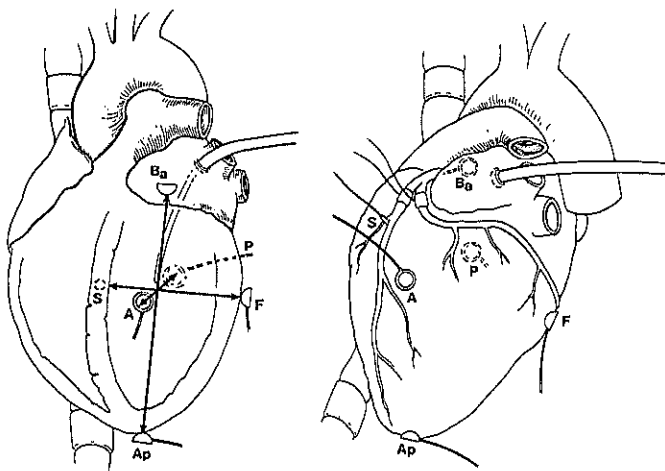


FIG. 1—*Left*, the basic preparation used to measure LV three-dimensional geometry with the LV anterior-posterior minor axis dimension (A to P), LV septal-free wall minor axis dimension (S to F), and the LV base-apex major axis dimension (Ba to Ap). Left atrial silastic intraducer is represented with a Millar monometer positioned within the LV. Pneumatic vena caval occluders are also represented around the superior and inferior vena cava. *Right*, left anterior oblique view of the LV showing the positions of the Doppler flow probes around the proximal LAD and circumflex coronary arteries. Remainder of the dimensions are the same as those depicted on the left.

Houston, TX) were placed around the left anterior descending coronary artery, distal to the first diagonal branch, and around the proximal circumflex coronary artery. After transducer placement, a silastic tube (2.6 mm i.d., 4.9 mm o.d., Dow Corning, Midland, MI) was secured in the base of the left atrial appendage for later passage of a micromanometer (Millar Instruments model PC-350, Houston, TX) into the left ventricle (LV). A similar silastic tube with multiple side holes was positioned in the pleural space near the LV epicardium for measurement of pleural pressure. The transducer leads, catheters, and occluder tubing were all exteriorized dorsal to the thoracotomy and placed in a subcutaneous pouch. The pericardium was left wide open and the thoracotomy was repaired in multiple layers. All dogs received i.m. injections of cephalexin (500 mg twice daily) for 5 days postoperatively. The dogs were allowed to recover for 7–10 days before the hardware was exteriorized through a small skin incision using 1% lidocaine local anesthesia.

Data acquisition and experimental protocol

One day after exteriorization, each dog was sedated lightly (morphine sulfate, 5–7 mg i.m.) 1 h before data acquisition. The right neck was anesthetized with 1% lidocaine local anesthesia, and using the Seldinger technique a 7-French introducer was placed into the internal jugular vein. Under fluoroscopic guidance, a catheter was advanced into the right ventricle and positioned into the coronary sinus (CS). Once placement was verified, the catheter was securely sutured into place and a 3-way stopcock was positioned on the end of the catheter so that CS blood could be routinely sampled. Each dog was then studied in the conscious state while lying on its right side in a large, multiplace hyperbaric chamber (F.G. Hall Laboratory for Environmental Research, Duke University).

The ultrasonic dimension transducers were coupled to electrical cables which passed from inside the chamber to outside the chamber by way of an airtight port and were then directly coupled to a sonomicrometer developed and constructed in

the laboratory of the Department of Surgery, Duke University. The sonomicrometer has a practical frequency response of 0–50 Hz with a minimal resolution of 0.08 mm and maximal electronic drift of 0.05 mm/h. The Doppler flow probes were coupled to cables inside the chamber, which were then directly coupled to a Doppler flow analyzer (Craig Hartley) outside the hyperbaric chamber. A silastic T-tube (with two ports) was placed on the end of the silastic tube which had been implanted in the left atrial appendage. A 3-way stopcock was placed on one port of the T-tube, which allowed for routine sampling of arterial blood. A micromanometer (Millar PC-350, Houston, TX) was passed through the other port into the LV. Another similar micromanometer was passed through the pleural silastic tube and positioned in the pleural space adjacent to the LV. The micromanometers were coupled to cables inside the chamber which were then directly coupled to pressure amplifiers (model 8085-L, Hewlett-Packard, Waltham, MA) located outside the hyperbaric chamber. Each micromanometer had been prewarmed (38°C), balanced, and calibrated against a water column before each study. Calibration was rechecked at the conclusion of each study, and resultant micromanometer drift was less than 0.5 mmHg in all studies. Blood gas determinations were made with an Instrumentation Laboratory pH and blood gas analyzer (model 113) inside the chamber, calibrated with known gases at the same pressure at which the samples were obtained. All determinations were done in duplicate and averaged. The details of this method have been described previously (6, 7). Analog data were digitized in real time at 200 Hz. The performance characteristics of these instruments have been described previously (8–10).

In each dog, pharmacologic attenuation of autonomic reflexes was accomplished before all measurements by i.v. administration of propranolol (1–2 mg/kg), phentolamine (0.5 mg/kg), and atropine (0.4 mg/kg). Measurements of LV pressure, pleural pressure, ultrasonic dimensional data, arterial and CS blood gases were made consecutively during control and transient vena caval occlusions (VCO) under the following conditions:

1. At 1 atm abs pressure while breathing air, 21% oxygen (1 bar/0.21).
2. At 3 atm, after 10–15 min for equilibration, while breathing compressed air, 21% oxygen (3 bar/0.21).
3. At 3 atm, after 10–15 min of breathing 100% oxygen (3 bar/1.0).
4. At 1 bar/0.21, after 30 min for equilibration, while breathing air.

An investigator and a technician were continuously present inside the chamber with the animal. Gas composition provided to the dog was changed by having the dog wear an air-tight gas-exchange hood sealed around the neck with a rubber collar. The inspired gas composition was continuously monitored through the gas-exchange hood to ensure that the proper gas mixture was being administered.

At the conclusion of all studies, the animals were euthanized, the hearts were excised, and the proper position of the hardware was verified. LV wall volume was calculated as the average of 3 vol displacements of water in a graduated cylinder after excision of the atria, right ventricular free-wall, aortic and mitral valves, and chordae tendineae.

Data analysis

Analysis of digitized data was performed on a microprocessor (DEC model, PDP 11/23, Maynard, MA) with interactive software developed in our laboratory. LV

transmural pressure was calculated as the difference between LV intracavitary pressure and the simultaneous pleural pressure. LV intracavitary volume (VL) was calculated from the LV epicardial major (a), anterior-posterior minor (b), and septal-free wall minor (c) axis diameters using a modified ellipsoidal shell model with subtraction of LV wall volume (VL_{wall}) (8-10). The formula for the volume of a modified ellipsoid is equivalent mathematically to that for a general ellipsoid:

$$VL = \frac{\pi \cdot a \cdot b \cdot c}{6} - VL_{wall}$$

where a is equivalent to segment length ApBa, b is equivalent the segment length AP, and c is equivalent to the segment length S-F in Fig. 1 *left*. LV stroke volume was computed as the change in the LV internal shell volume during ejection. CO was calculated as the product of LV stroke volume and the simultaneous heart rate (HR).

The LV stroke work (SW) was computed as the integral of the respective LV transmural pressure (P) - volume (V) loop over each cardiac cycle:

$$SW = \int P \cdot dV$$

The relationship between SW and end-diastolic volume (EDV), the pre-load recruitable stroke work (PRSW), during transient vena caval occlusion was determined by linear-regression analysis as described previously (11), and data were fitted to the formula:

$$SW = M_w(EDV - V_w)$$

where M_w and V_w are the slope and X intercept, respectively.

Total CBF (TCBF) was computed from the sum of the individual flows through the proximal left anterior descending (LAD) and the proximal circumflex (CX) coronary arteries:

$$TCBF \text{ (ml/s)} = LAD \text{ (ml/s)} + CX \text{ (ml/s)}.$$

Arterial (AO₂ content) and CS (CSO₂ content) oxygen contents were determined from their respective oxygen saturation and oxygen tensions as follows:

$$O_2 \text{ CT} = [\text{Hgb}] [\% \text{ Sat}] [1.34] + PO_2 (0.003)$$

where O₂ CT is the oxygen content (either AO₂ or CSO₂); Hb is the determined Hb; % Sat is the percentage of oxygen bound to Hb; 1.34 represents a constant which relates the amount of oxygen bound to a gram of hemoglobin; and PO₂ is the oxygen tension (either PaO₂ or CSPO₂). The quantity PO₂ (0.003) represents that portion of oxygen which is dissolved in the plasma.

The arterial-venous oxygen difference (A-CSO₂) was calculated as the difference between the AO₂ content and the CSO₂ content:

$$A-CSO_2 = AO_2 \text{ content} - CSO_2 \text{ content}$$

Myocardial oxygen consumption (MVO_2) was determined from A-CSO₂ and TCBF:

$$\text{MVO}_2 = (\text{A-CSO}_2) \times \text{TCBF}$$

This value is presented in units of joule per beat per 100 g of myocardium by converting from milliliter O₂ to joule with the following equation:

$$1 \text{ ml O}_2 = 20.1 \text{ J}$$

Comparisons of cardiac dimensions, volume, pressure, arterial and CS blood gases, as well as MVO_2 data were made between 1 bar while breathing 21% O₂ (1 bar/0.21), 3 bar while breathing 21% O₂ (3 bar/0.21), and 3 bar while breathing 100% O₂ (3 bar/1.0). Statistical comparisons were performed with multiple analysis of variance. All results are expressed as mean $\pm$ SD.

RESULTS

Individual and mean cardiac dimension, stroke volume, HR, and output and pressure data obtained during control (1 bar/0.21), 3 bar/0.21, and 3 bar/1.0 are presented in Table 1. For all 10 animals, mean hemodynamic data demonstrated no significant changes in HR, mean LV ejection pressure, LV end-diastolic pressure, pleural pressure, or LV EDV. Mean LV stroke volume showed no significant change between control and 3 bar/0.21; however, it was significantly reduced during 3 bar/1.0. Stroke volume fell from 24.6 ± 4.8 ml during control to 20.2 ± 8.2 ml during 3 bar/1.0 ($P < 0.05$). CO fell from 2.7 ± 0.7 liter/min at control to 2.5 ± 0.6 liter/min at 3 bar/0.21 ($P < 0.01$), and to 2.1 ± 0.7 liter/min at 3 bar/1.0 ($P < 0.01$ vs. control and 3 bar/0.21).

Representative LV pressure-volume loops for each of the three conditions are shown in Fig 2. The three loops depicted are essentially equivalent. Even though the LV EDV decreased from control to 3 bar/1.0 and the peak systolic LV transmural pressure increased from control to 3 bar/1.0, the area contained within the three loops (the SW) is essentially the same. This relationship was noted in all 10 experimental animals. To further quantify myocardial performance, LV SW (area under the pressure-volume loop) was plotted as a function of LV EDV. This relationship is consistently linear. A representative SW-EDV relationship (PRSW) is depicted in Fig 3, and the individual slopes and X-intercepts are shown in Table 2. For all 10 animals the PRSW was not significantly different between any of the three conditions (Table 3), indicating that myocardial contractility was preserved.

Mean TCBF, myocardial oxygen consumption, and arterial and CS blood gases are presented in Table 3. Pa_{O_2} increased from 85.7 ± 5.1 mmHg at control to 324.6 ± 229 mmHg at 3 bar/0.21 ($P < 0.01$), and to 1374 ± 201 mmHg at 3 bar/1.0 ($P < 0.01$ vs. control). Pa_{CO_2} increased from 39.3 ± 8.1 mmHg at control to 44.1 ± 10.7 mmHg at 3 bar/0.21 ($P < 0.05$) and was not significantly different from control at 3 bar/1.0 (41.9 ± 13.3 mmHg). Arterial pH was not significantly different between any of the three conditions. AO_2 content, and CSO₂ content both increased significantly from control to 3 bar/0.21 and to 3 bar/1.0. The A-CSO₂ difference was significantly smaller at 3 bar/0.21; however, this variable was not significantly different at 3 bar/1.0. The Pa_{O_2} , AO_2 content, and CSPO₂ were all significantly higher at 3 bar/1.0 when compared to 3 bar/0.21.

Table 1: Individual and Mean Left Ventricular Hemodynamic Pressure and Volume Data

Dog No.	HR, beats/min			EDP, mmHg			EJP, mmHg			PLP, mmHg			EDV, ml			SV, ml			CO, liter/min		
	1/0.21	3/0.21	3/1.0	1/0.21	3/0.21	3/1.0	1/0.21	3/0.21	3/1.0	1/0.21	3/0.21	3/1.0	1/0.21	3/0.21	3/1.0	1/0.21	3/0.21	3/1.0	1/0.21	3/0.21	3/1.0
1	151.9	148.8	151.9	12.0	14.1	13.6	125.4	117.1	116.1	-3.3	-4.0	-4.1	45.3	44.8	43.7	21.1	19.1	16.5	3.2	2.8	2.5
2	111.8	107.7	107.2	16.9	14.9	14.7	112.4	113.2	120.9	-4.2	-4.7	-5.5	40.4	40.8	39.9	16.3	15.6	14.8	1.8	1.7	1.6
3	127.2	125.3	123.9	6.4	7.3	4.5	140.0	145.5	143.4	-2.0	-2.3	-0.7	40.0	39.7	39.0	31.2	29.4	26.2	4.0	3.7	3.2
4	101.7	81.0	75.0	10.2	10.7	11.0	121.5	120.3	118.7	-5.3	-6.7	-6.1	62.9	70.2	71.9	34.8	38.2	40.2	3.5	3.1	3.0
5	88.5	90.6	86.9	14.9	7.6	9.7	113.3	89.0	89.9	-5.1	-3.5	-4.0	50.3	38.9	41.9	19.0	17.8	18.8	1.7	1.6	1.6
6	91.1	92.0	91.2	8.5	8.7	8.5	100.1	118.3	119.5	-1.4	-2.1	-2.7	42.7	39.1	35.6	22.1	20.2	12.7	2.0	1.9	1.2
7	97.7	100.5	102.6	11.9	10.1	8.7	96.6	124.3	124.9	-1.4	-2.2	-0.9	53.9	35.2	30.4	25.4	23.6	11.7	2.5	2.4	1.2
8	132.2	128.2	125.5	11.5	13.4	19.2	114.9	120.0	108.9	-1.5	-3.1	-3.1	43.6	44.5	47.6	24.2	22.8	20.4	3.2	2.9	2.6
9	105.0	91.8	90.6	18.2	17.9	19.5	118.9	129.4	131.3	-5.8	-4.7	-5.1	70.5	78.1	79.7	28.3	27.5	25.9	3.0	2.5	2.3
10	110.0	111.0	109.1	5.9	8.8	7.7	111.5	114.0	112.6	-1.1	-2.1	-0.7	45.7	45.2	46.1	22.9	20.1	15.7	2.5	2.2	1.7
Mean	111.7	107.7	106.4	11.7	11.4	11.7	115.5	119.1	118.1	-3.1	-3.5	-3.3	49.5	47.7	47.6	24.6	23.4	20.2 ^c	2.7	2.5 ^a	2.1 ^a
±SD	18.9	20.0	21.5	3.9	3.4	4.7	11.7	13.4	13.8	1.8	1.4	1.9	9.6	13.7	15.0	4.8	6.4	8.2	0.7	0.6	0.7
P	>0.05			-0.10			>0.10			-0.10			-0.05			-0.10			<0.01		

Key: HR: heart rate; EDP: left ventricular end-diastolic transmural pressure; EJP: mean left ventricular ejection transmural pressure; PLP: mean pleural pressure; EDV: left ventricular end-diastolic volume; SV: left ventricular stroke volume; CO: cardiac output. 1/0.21: control data, 1 atmosphere pressure while breathing 21% oxygen (room air). 3/0.21: 3 atmospheres pressure while breathing 21% oxygen. 3/1.0: 3 atmospheres pressure while breathing 100% oxygen. $P < 0.05$ vs. control 1/0.21

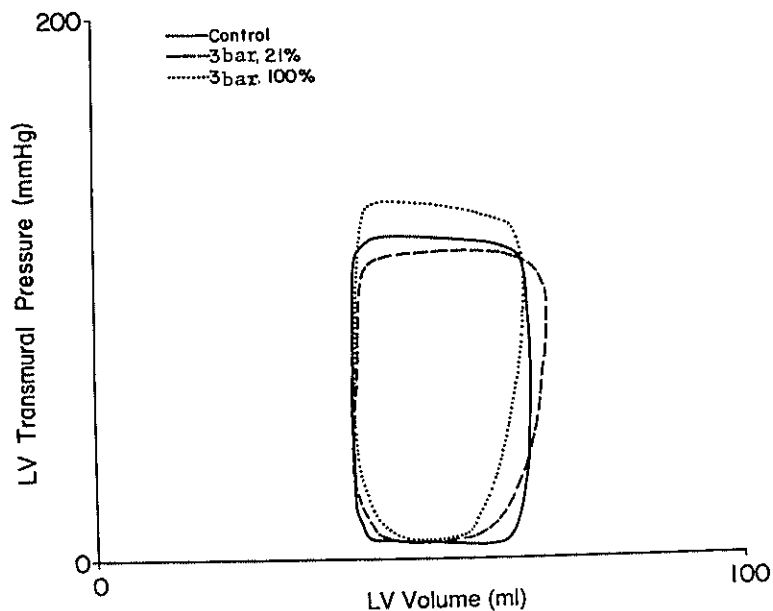


FIG. 2—Representative LV pressure-volume loops for control (solid line), 3 bar while breathing air (broken line), and 3 bar while breathing 100% oxygen (dotted line).

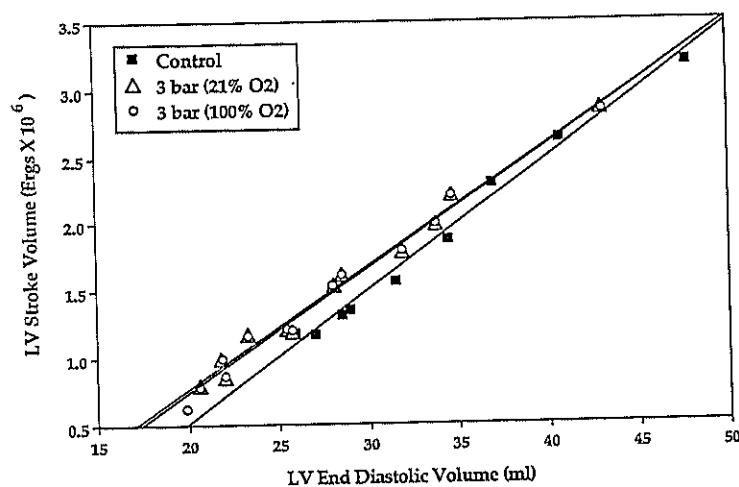


FIG. 3—Representative linear SW-end-diastolic volume relationships (PRSW) for control, 3 bar while breathing compressed air, and 3 bar while breathing 100% oxygen.

With increasing P_{aO_2} TCBF decreased (Table 2). TCBF was reduced from 97.7 ± 8.0 ml/min at control to 89.6 ± 11.7 ml/min at 3 bar/0.21 ($P < 0.05$), and to 85.3 ± 13.6 ml/min at 3 bar/1.0 ($P < 0.05$ vs. control and 3 bar/0.21). With decreasing TCBF MVO_2 also decreased (Table 3). The MVO_2 fell from 1.30 ± 0.3 J $\cdot$ beat $^{-1} \cdot 100$ g $^{-1}$ (myocardium) at control to 1.20 ± 0.3 J $\cdot$ beat $^{-1} \cdot 100$ g $^{-1}$ at 3 bar/0.21 ($P < 0.05$),

Table 2: Mean Total Coronary Flow, Myocardial Oxygen Consumption, Arterial Blood Gases, and Coronary Sinus PO₂ Data

	TCBF, ml/min	Paco ₂	pH	Pao ₂	Ao ₂ Content, vol%	VCSO ₂ Content, vol%	A-CSO ₂ , vol%	CSPo ₂	MV _{O₂} , l/b/100g
1 bar 0.21 O ₂	97.7 ±8.0	39.3 ±8.1	7.287 +0.9	85.7 ±5.1	13.2 ±0.3	4.7 ±0.9	8.3 ±1.0	25.7 ±3.1	1.30 ±0.3
3 bar 0.21 O ₂	89.6 ±11.7	44.1 ±10.7	7.279 ±0.9	324.6 ±22.9	14.0 ±0.5	9.1 ±1.9	4.9 ±1.8	32.3 ±4.1	1.20 ±0.3
P vs. control	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05
3 bar 1.0 O ₂	85.3 ±13.6	41.9 ±13.3	7.311 ±0.12	1374 ±201	17.0 ^a ±0.7	10.1 ±1.3	7.1 ±0.6	41.4 ^a ±9.2	1.10 ±0.3
P vs. control	<0.05	>0.20	>0.05	<0.05	<0.05	<0.05	>0.10	<0.05	<0.05

Key: TCBF: Total coronary blood flow; Paco₂: arterial CO₂ tension; pH: arterial pH; Pao₂: arterial O₂ tension; Ao₂ content: arterial oxygen content; VCSO₂: coronary sinus oxygen content; A-VCSO₂: arterial-venous oxygen difference; CSPo₂: coronary sinus oxygen tension-MVO₂; Myocardial oxygen consumption. The remainder of the variables are the same as in Table 1.

^aP < 0.05 for 3/1.0 data vs. 3/0.21

Table 3: PRSW Relationship Slopes and X-intercepts Under Control Conditions and at 3 bar While Breathing 21% Oxygen and 100% Oxygen

Dog No.	Control: 1 bar -0.21 O ₂				3 bar -0.21 O ₂				3 bar -1.0 O ₂			
	Mw, ergs × 10 ⁵	Vw, ml	r		Mw, ergs × 10 ⁵	Vw, ml	r		Mw, ergs × 10 ⁵	Vw, ml	r	
1	1.15	13.9	0.973		1.14	14.8	0.955		1.13	19.7	0.949	
2	1.07	31.5	0.977		1.09	32.6	0.995		1.13	32.2	0.958	
3	1.51	2.4	0.985		1.49	0.3	0.994		1.45	4.4	0.975	
4	0.813	8.6	0.989		0.819	6.6	0.989		0.752	-0.9	0.979	
5	1.07	12.0	0.983		0.740	2.4	0.993		0.807	2.9	0.985	
6	1.30	59.4	0.982		1.42	43.0	0.983		0.961	43.0	0.971	
7	0.963	14.6	0.999		0.913	16.4	0.998		0.919	21.6	0.974	
8	0.989	4.9	0.995		0.919	1.8	0.993		0.915	1.7	0.997	
9	0.684	29.8	0.983		0.702	33.2	0.976		0.724	32.8	0.980	
10	1.09	7.2	0.985		1.06	6.9	0.988		1.01	8.0	0.970	
Mean	1.06	18.4	0.985		1.03	15.8	0.986		0.981	16.5	0.974	
± SD	±0.22	±16.5	±0.007		±0.25	±14.5	±0.012		±0.21	±14.7	±0.012	
P value vs. control						>0.50				>0.20		

Key: Mw: slope of the PRSW relationship; Vw: X-intercept of the PRSa relationship; r: linear-correlation coefficient for the PRSW relationship.

and to $1.10 \pm 0.3 \text{ J} \cdot \text{beat}^{-1} \cdot 100 \text{ g}^{-1}$ at 3 bar/1.0 ($P < 0.05$ vs. control). The fall in MVO_2 from 3 bar/0.21 to 3 bar/1.0 did not reach statistical significance. All values returned to baseline at the completion of each study, when the dogs were restudied at 1 bar/0.21.

DISCUSSION

Clinical interest in HBO is recent, but scientific curiosity about high-pressure environments dates back to antiquity (12). Over the past two centuries, scientists in many countries have studied the effects of increased environmental pressures on biological systems; however, no firm data as to the physiologic effects of both pressure and oxygen on the cardiac system have been established. In the early 1950s, Boerema and associates (11) reintroduced the hyperbaric chamber into the modern medical era by demonstrating enhanced oxygenation when increased atmospheric pressures were combined with the respiration of pure oxygen. This prompted some cardiac surgeons, particularly in Europe and the Soviet Union, to experiment with performing cardiac operations, primarily for cyanotic heart disease, under the influence of HBO (1, 13). Initial studies also suggested a prolonged survival in animals with acute myocardial infarction while breathing oxygen under increased pressure (14–16). Patients with coronary artery disease were reported to have an increased exercise tolerance and a decreased incidence of ventricular arrhythmias while being treated with HBO. As experimentation progressed, mixed results surfaced as to the benefits of HBO in cardiac surgery and in the treatment of coronary occlusive disorders (2, 12, 13, 17, 18). Studies revealed that there was a decrease in CO, CBF, HR, and myocardial oxygen delivery during HBO (6, 12, 13). These controversies contributed to HBO becoming almost nonexistent in the primary treatment protocol of patients with cardiac disease. An increasing number of critically ill patients are treated with HBO. It is important that clinicians understand the cardiovascular responses to HBO, because a large percentage of these patients have some degree of cardiovascular impairment.

The present study was undertaken to incorporate modern techniques for quantifying myocardial contractility and blood flow in an attempt to better understand the varied results obtained in the past. The geometric assumptions and limitations of the model used in this study (the modified ellipse) have been discussed in detail elsewhere (8–10). The PRSW has been shown to be a load-insensitive index of myocardial contractility (9). No model used in the previous studies on this subject has satisfied this important requirement (4, 6, 12, 19–21). As demonstrated in Fig. 2, HBO results in alterations of both LV preload and LV afterload. These alterations account for many of the conflicting results observed in the previous literature. The PRSW, which has been shown to be independent of both preload and afterload, was utilized in the present study.

The circulatory changes that occur during HBO result from the complex interaction of oxygen directly on the heart and vascular bed as well as its indirect effects mediated through neurohormonal mechanisms. The large number of different variables that interact with the cardiovascular system often makes interpretation of pure myocardial properties difficult. To properly evaluate the effects of HBO on cardiac contractility and oxygen consumption, and to make direct comparisons between control and

increasing levels of oxygenation, HR must remain constant. Changes in HR result in direct alterations in cardiac contractility (9) as well as MVO_2 (22, 23). Many of the previous investigations have neglected to control HR and have therefore made it impossible to determine the mechanism responsible for the changes seen in cardiac performance (3, 6, 13, 20, 24–26). To separate these variables, autonomic blockade was utilized in the present study. Autonomic blockade has been shown to eliminate the hyperbaric bradycardia that is often observed (3, 6, 12, 13, 20, 24). This was confirmed in the present study since no significant difference in HR occurred (Table 1).

The present communication demonstrates no significant reduction in myocardial contractility with increasing PaO_2 . This result was similar to the finding by Ishikawa et al. (5): In 34 dogs the left descending coronary artery was directly perfused with oxygenated blood (463 mmHg) which resulted in an increase in myocardial contractile force by $11.1 \pm 2.3\%$. If the animals inhaled 100% O_2 while the LAD continued to be perfused with oxygenated blood, there was a $4.1 \pm 2.1\%$ reduction in myocardial contractile force, indicating a return of myocardial contractility toward control levels. Myocardial contractility was preserved even with a significant reduction in CBF, LV, stroke volume, and CO. These observations are in accordance with the previous body of literature (20, 21, 23, 24, 27, 28). This indicates that local vasoconstriction must occur with increasing PaO_2 . Decreasing CO is indicative of systemic vasoconstriction, whereas decreasing CBF is a direct result of diffuse coronary vasoconstriction. Local autoregulation, probably at the capillary level, is most likely responsible for this finding. Increased oxygen tensions (above normal) are presumably toxic to tissues, therefore suggesting that local autoregulation decreases blood flow in an attempt to maintain normal (or control) levels of oxygen delivery. In the myocardium the decrease in CBF does not seem to result in altered myocardial performance because contractility is preserved. Vasoconstriction, most likely, is one of the normal regulatory or protective responses of the myocardium and peripheral tissues (and possibly visceral organs as well) to an increase in oxygen tension above normal.

Myocardial oxygen consumption decreased with increasing levels of oxygenation. This is in accordance with various other studies (23, 25). Ishikawa et al. (23) were able to demonstrate a reduction in MVO_2 with a concomitant 10% reduction in LV tension–time index and a statistically insignificant decline in Max dp/dt in patients with coronary artery disease, breathing 100% oxygen. The investigators hypothesized that oxygen breathing diminishes myocardial contractility and consequently may reduce myocardial oxygen demand. The present investigation is unique in that there was a decline in MVO_2 without any reduction in contractility. Therefore, the reduction seen in MVO_2 is not related to decreased work of the myocardium. The heart is an aerobic organ, so it relies almost exclusively on oxidation of substrates for the generation of energy, and it can develop only a small oxygen debt. Since the heart relies almost exclusively on oxidative metabolism for energy generation, the rate of cardiac oxygen consumption can provide a useful index of its total energy utilization (23). The oxygen demand of the myocardium is related to the energy required for contractile activities as well as the basal and activation metabolism for non-contractile activities (22). MVO_2 varies directly with the fraction of energy derived from the metabolism of fatty acids, which in turn varies directly with the arterial concentration of fatty acids and inversely with that of insulin and glucose (29). Since the reduction in MVO_2 does not seem to be related to a reduction in the energy required for

contractile activities, it must be a result of a reduction in the energy requirement for non-contractile activities. By definition, the basic energy requirement of the myocardium must remain constant; therefore the reduction in MVO₂ must be due to a decrease in energy metabolism for various substrates. This concept is supported by studies on the effects of oxygen on the perfusion and metabolism of the ischemic myocardium (26). It has been demonstrated that inhaling 100% O₂ results in a decrease in myocardial lactate extraction in both dogs and man (18, 26). Decreased lactate extraction could be indicative of a decreased production of lactate secondary to improving ischemia or decreased utilization or delivery of substrates vital in the production of lactate. Therefore, decreased MVO₂ could be indicative of a) acute oxygen toxicity with mitochondrial damage and decreased oxygen utilization, or b) more efficient substrate utilization and metabolism in an oxygen rich environment, or c) decreased substrate availability secondary to vasoconstriction. The present investigation, however, does not help to select any of these alternative hypotheses.

SUMMARY

The effects of increased oxygen partial pressure on the myocardium involves an intricate array of local and neurohormonal factors. The present study utilized pharmacologic autonomic blockade to eliminate some of the neurohormonal factors responsible for alterations in the cardiovascular response to increased PaO₂. This allowed for a clearer separation between the direct and secondary effects of oxygen on the myocardium. The present study indicates that increased PaO₂ has the primary effect of reducing MVO₂ whereas myocardial contractility is preserved. Increased PaO₂ resulted in a decreased LV stroke volume and CO (without a change in HR) as well as a reduction in TCBF. This indicates that diffuse vasoconstriction is most likely a locally mediated autoregulatory response of the myocardium and peripheral tissues to an increase in PaO₂. Vasoconstriction protects the myocardium and peripheral tissue from possible toxic effects of increased oxygen. It is important to point out that this occurs in the myocardium without any alteration in contractility.

The exact mechanism responsible for the reduction seen in MVO₂ remains elusive. However, because contractility was preserved, the energy required (or oxygen consumed) for contractile activities must have remained constant. Therefore, the reduction in MVO₂ must indicate a decrease in the energy utilized for non-contractile activities. The energy required for non-contractile activities includes the basal energy expenditure and the energy required for the utilization of substrates in the activation of metabolism. This could indicate: a) the onset of acute oxygen toxicity, b) more efficient substrate utilization or metabolism or both, or c) decreased substrate availability secondary to vasoconstriction; or a combination of the above processes. It will be prudent for future investigators to sort out the exact mechanism responsible for this reduction in MVO₂ because this may lead to future alterations in the therapeutic protocol for various myocardial diseases. Decreased MVO₂ secondary to decreased substrate availability or more efficient metabolism can possibly be helpful during the reperfusion phase of myocardial ischemia. This could lead to a reduction in the production of toxic metabolites in a previously ischemic area of myocardium, which in turn could lead to a reduction in reperfusion injury.

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